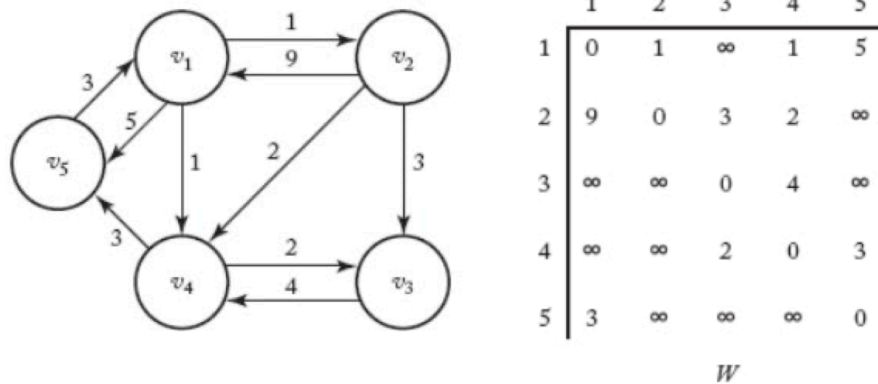


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 # Description: Homework Assignment 4

- Using Floyd's shortest path problem discussed in class (and described on Pg 100 of the textbook), compute terms $D^{(1)}[2][3]$, $D^{(1)}[5][2]$, $D^{(1)}[5][3]$, and $D^{(2)}[5][3]$ for the $W = D^{(0)}$ given below.



For this, you would need to use the following recursive property. If you need an example to help you, please see Example 3.3 on Page 106 of the textbook.

$$D^{(k)}[i][j] = \text{minimum} \left(\underbrace{D^{(k-1)}[i][j]}_{\text{Case 1}}, \underbrace{D^{(k-1)}[i][k] + D^{(k-1)}[k][j]}_{\text{Case 2}} \right)$$

$$\begin{aligned} D(1)[2][3] &= \min D(0)[2][3], D(0)[2][1] + D(0)[1][3] \\ &= \min 3, 9 + \infty \\ &= \min 3 \end{aligned}$$

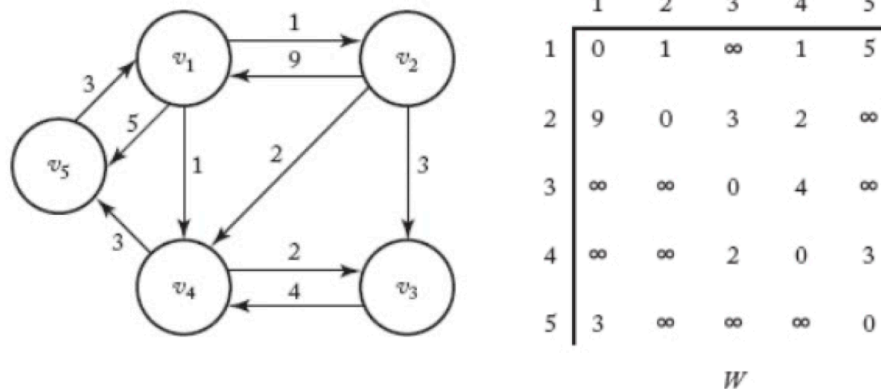
$$\begin{aligned} D(1)[5][2] &= \min D(0)[5][2], D(0)[5][1] + D(0)[1][2] \\ &= \min 4, 3 + 1 \\ &= \min 4 \end{aligned}$$

$$\begin{aligned} D(1)[5][3] &= \min D(0)[5][3], D(0)[5][1] + D(0)[1][3] \\ &= \min \infty, 3 + \infty \\ &= \min \infty \end{aligned}$$

$$\begin{aligned} D(2)[5][3] &= \min D(1)[5][3], D(1)[5][2] + D(1)[2][3] \\ &= \min \infty, 4 + 3 \\ &= \min 7 \end{aligned}$$

- Implement Floyd's algorithm to find the length of the shortest path in Python. Use the following graph and its corresponding W matrix as the input – you can use a 2D list to

store the values in the matrix. You can use a large number like 10000 to represent infinity in the matrix. Your output should print the starting matrix $W = D^{(0)}$, and all the intermediate matrices $D^{(1)}$ through $D^{(4)}$ and the final matrix $D = D^{(5)}$. Please use Algorithm 3.3 on page 106 of the textbook to guide your implementation.



**** Code is implemented through github, also will zip the folder ****

3. Suppose we have the following six matrices with their respective dimensions shown below:

$$\begin{array}{cccccc}
 A_1 & \times & A_2 & \times & A_3 & \times & A_4 & \times & A_5 & \times & A_6 \\
 5 \times 2 & & 2 \times 3 & & 3 \times 4 & & 4 \times 6 & & 6 \times 7 & & 7 \times 8 \\
 d_0 & d_1 & d_1 & d_2 & d_2 & d_3 & d_3 & d_4 & d_4 & d_5 & d_5 & d_6
 \end{array}$$

Based on the material covered in class on the dynamic programming approach on chained matrix multiplication, we know the following recursive property holds for items in the array M:

$$M[i][j] = \underset{i \leq k \leq j-1}{\text{minimum}} (M[i][k] + M[k+1][j] + d_{i-1}d_kd_j), \text{ if } i < j.$$

$$M[i][i] = 0.$$

- a) Using the dynamic programming approach discussed in class, compute the following terms in the array M shown in Figure 3.8 on page 115 of the textbook. As shown in class, please show steps on how each term is computed (see example 3.6 on page 119 for guidance). You may use values from previous diagonals as needed from the M matrix in the Figure 3.8 on page 115 of the textbook. You can also double-check your answer using Figure 3.8:

$$M[3][3]$$

$$M[3][4]$$

$$M[3][5]$$

$$M[3][6]$$

$M[1][5]$

$M[1][6]$

b) Based on the way calculations were done above, briefly explain in your own words how this is a bottom-up approach to solving the problem.

 MatrixMul.pdf